

Vimalkumar Patel

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Senior/Staff Software Engineer ready to accelerate in a challenging environment with 15+ years of experience building platform infrastructure that scales privacy and compliance operations across a 100K+ employee organization. Architected Agentic AI Workflows, automated scheduling, SLA enforcement, and work distribution systems serving 78K+ users. Track record of driving 90%+ performance improvements, eliminating thousands of manual operational hours, and leading cross-functional initiatives spanning engineering, product, UX research, and program management.

Skills

Programming: Java, PHP/Hack, Python, C/C++, JavaScript/TypeScript, React, GraphQL, SQL, Spring Boot, JUnit, Hibernate/JPA, SQL/NoSQL, Shell

Tools: Docker, AWS, GIT, Maven, SonarQube, Splunk, Postman, Mockito, WireMock, Jenkins, Jira, Swagger

Practices: System design, API architecture, SLA modeling, capacity planning, performance engineering, technical leadership

Misc: RESTful API, Microservices, Design Patterns, TDD, Agile Methodologies, Certified MongoDB Developer

Education University of Illinois at Chicago | Master of Science - Computer Science | May 2014

Experience

Meta, NYC | Senior Software Engineer | Aug 2019 – Present

Platform Infrastructure & Regulatory Inquiry Tooling (2024–2026)

- Deployed Metamate AI agents to automate business-critical, high-touch Regulatory Inquiry Response workflows, **reducing manual response processing time to <15 minutes** per inquiry
- **Designed and engineered** production-grade LLM observability dashboards with a continuous evaluation pipeline.
- Implemented drift detection for Regulatory Inquiry AI agents, utilizing a hybrid topology of code-based and LLM-as-a-Judge graders to ensure high-fidelity, regulatory-compliant responses.
- **Engineered audited ground-truth datasets** with granular access controls and optimized preprocessing; unlocked automated evaluations, prompt optimizations, and LLM analytics that **slashed latency/costs while delivering a 2X improvement in dataset quality**.
- Designed and Shipped support for parallel multi-file processing and abstracted file-format handling; **expanding end-to-end agentic workflow coverage by 29%** across highly complex, non-standard enterprise payloads.
- **Introduced advanced runtime optimizations**—including dynamic model configuration, concurrency controls, and robust SLI integrations—to fortify the reliability and throughput of agentic workflows.
- **Led the end-to-end test environment enablement** across multiple toolsets; established defensive data boundaries by **deprecating legacy workflows polluting production environments**.
- **Architected and delivered a unified, action-oriented review inbox** for regulatory inquiries, driving high-impact engagement that **surpassed strategic targets with over 50% of inquiries resolved within 14 days**.
- **Spearheaded a comprehensive platform performance modernization initiative**, capturing corporate recognition for engineering excellence by **slashing page latency by 93.8% (9.3s → 580ms)**, **reducing memory footprints by 87%**, and **mitigating user-impacting defects by 93.3%**.
- **Championed the organizational "Engineering Excellence" maturity initiative**, upskilling engineering teams and formalizing production-ready standards across platform quality, reliability, and observability frameworks.
- **Authored the long-term strategic technical vision and roadmap** for insights tooling, establishing architectural directions and high-priority backlogs actively adopted by cross-functional teams to execute deliverables.

Workflow Automation & Operational Efficiency (2023–2024)

- **Architected an enterprise operations platform** for bulk compliance management; drove **100% organizational adoption** and eliminated 52+ hours of manual overhead monthly for program operations.
- Redesigned company-wide task reassignment workflow **slashing annual operational overhead from thousands of hours to ~400 hours**, and **reducing per-manager effort from hours to <2 minutes** per request.
- Designed extensible notifications framework with cross-system tracking achieving 30%+ engagement rates; 6.7% improvement in assignment ownership confirmation and 3.8% higher task completion in pilot.
- Automated recurring proposal generation eliminating manual creation of 49+ proposals per cycle

Task SLA Enforcement & Ownership Systems (2022–2023)

- Architected and rolled out **SLA commitment** system as a result;

Impact: tasks reached correct owner 3 days faster, and advanced first-action timelines by **5 days**.

- Drove **ownership accuracy initiative** via cross-functional partnership: achieved ~530 hours savings in task transfer time every month; improved time-to-last-owner by 8%
- Defined and Implemented **federation contracts** enabling multi-track parallel work execution on the core platform
- Directed cross-functional technical synchronization with UX researchers, TPMs, and partner engineering teams through iterative design cycles

Scheduling & Capacity Planning Systems (2021–2022)

- Architected company-wide automated scheduling system that replaced manual asset processing and compliance task creation: **eliminated the operational overhead of 2 dedicated on-call engineers** per shift; Provided **7-day advance visibility** into work assignments for 2,300+ engineers
- **Designed a multi-constraint capacity dispatch engine** that dynamically balanced team/individual availability, priority queues, and strict SLA constraints.
- Engineered automated Impact Assessment Reports, replacing legacy spreadsheets with unified report providing organizational-level Level of Effort (LOE) breakdowns and predictive hotspot detection. **Provided absolute technical direction for the domain**, authoring core architecture specifications, executing multi-phase implementation breakdowns, and presiding over internal technical review boards.

Work Distribution Platform & Developer Tools (2019–2021)

- Core engineer on **Mobilizer/federation platform** managing 67K+ compliance tasks per cycle across the company
- Built **queue management system** and optimized the e2e system with 95%+ query performance improvement (28s → 800ms); optimized memory footprint by 90%
- Designed **task ownership workflows** capturing 61.1% adoption rate and 47K+ confirmations; eliminated manual triaging overhead entirely
- Compressed deployment package by 87.5% (4GB → 500MB), and Optimized test execution time by 86.4% and memory footprint by 68.1%

Engineering Excellence

- **AI adoption:** Top 2% company-wide in AI tool utilization; built AI-powered meeting recap features
- **Led a master database optimization initiative**, Compacted storage by 89% (20GB→2.2GB); reduced table reads by 99.76%; streamlining the total database footprint by **30%** (403GB→281GB)
- **Optimized caching layers** for shared UI components, reducing GraphQL network response latencies by 92.8%
- **Fortified system reliability** by resolving 4 high-severity production incidents; defined formal SLIs/SLOs for 3 product surfaces; implemented automated monitoring and alerting

People & Technical Leadership

- Managed and mentored 4 interns end-to-end - structuring technical project roadmaps, guiding execution, and leading calibration cycles.;
- Onboarded 5+ senior engineers across teams; served as cross-team buddy for 3 new hires, contributed to hiring interviews
- **Established an internal technical community**, hosting **15+ monthly technical brown bags** focused on Agentic AI systems, extreme performance tuning, and modern architectural patterns.
- **Galvanized engineering culture** by coordinating organizational summits, leading and winning corporate hackathons, and initiating cross-team peer recognition programs across multiple departments.

Target, San Francisco, CA | Lead Software Engineer | Consensus Retail Activation Platform | May 2019 – Aug 2019

- Provide technical leadership on API design, integration and testing with Spring cloud contracts.
- Led code quality initiatives and collaborated with architects on system design decisions, development processes, and introducing best practices using tools like Jacoco and SonarQube.

Wells Fargo, San Francisco, CA | Senior Software Engineer | AI Chatbot, Finance and Business Services | Nov 2017 – May 2019

- Architect Innovative enterprise AI solutions using Java, Spring boot, Deep learning, and advanced NLP libraries.
- **Designed system integration layer** using the Spring MVC framework, orchestrating secure external API connections to embed core banking systems into conversational AI platforms.
- **Engineered a 12.5x performance optimization** for a core chatbot platform, **slashing latency from 5+ seconds to 400ms** by implementing a low-latency in-memory database and an asynchronous batch-backup transcript architecture.

- Slashed intent resolution time by 50% across the conversation by designing and integrating an intelligent, high-throughput caching mechanism
- **Authored and standardized robust RESTful API specifications** using YAML and Swagger; established the integration fabric connecting internal functional engines with external vendors and downstream UI components.
- **Led the extensibility model for the conversational platform**, creating a modular framework to seamlessly inject new machine learning models, intent mapping, and multi-domain banking APIs.

Capital One, San Francisco, CA | Senior Software Engineer | Online Account Opening | Mar 2016 – Oct 2017

- executed a high-velocity test modernization strategy to expedite development, and scaling code coverage from 45% to 90% in 2.5 weeks using **Junit, Mockito, WireMock 1.57** and **SonarQube**.
- Designed and developed a POC to decompose legacy monolithic systems into a decoupled, Micro-service topology using Spring Boot.
- Drafted a **legacy modernization blueprint**, containerizing applications via Docker to transition workload orchestration from Oracle WebLogic 10 environments to modern Spring Boot microservices.
- **Orchestrated a major database modernization initiative**, leveraging the AWS Schema Conversion Tool to seamlessly migrate transactional schemas from legacy **Oracle 11g to cloud-forward PostgreSQL**.
- **Designed and scale-tested enterprise integration layers with external validation vendors**, driving a **33% increase** in new customer **automated approvals** through optimized identity matching pipelines.
- **Hardened application security and legal compliance** by re-architecting RESTful API workflows to strictly enforce Know Your Customer (KYC) regulatory frameworks.
- **Overhauled enterprise Single Sign-On (SSO) integration fabrics** with modern downstream identity APIs, significantly optimizing authentication funnels and reducing end-user drop-off rates.
- **Integrated a reliable event-driven audit and logging framework**, using distributed JMS queues, SLF4J, and Splunk to capture cross-layer systemic errors and audit-trail compliance events.
- **Devised data integrity validation POCs**, implementing automated cross-referencing against internal Systems of Record (SOR) using Python 3, ReadyAPI, and SmartyStreets.

Wells Fargo, San Francisco, CA | Java Developer | Shared Enrollment Business Services | Nov 2014 – Mar 2016

- Designed and integrated REST APIs and XDS interfaces across Wells Fargo Products using **Spring-MVC, JMS messaging APIs**, and BPM Tools.
- Developed Complex SQL and PL/SQL database Stored Procedures, Functions, Packages and Triggers.

Yelp Inc., San Francisco, CA | Software Programmer | Security & Privacy | July 2014 – Oct 2014

- Detection of exploitable bugs, and PII leakage in the source code & logs, and Mitigation via appropriate patches

CME Groups, Chicago, IL, USA | Student Intern | QA Tools and Automation | May 2013 – Aug 2013

- Development & optimization of order processing module to support high volume Post Trade Technology at CME
- Implemented automated Dashboard tool for mapping requirements to test cases, input data, and test scenarios across various platforms including Jira, Confluence, Quality Center, and GIT.

Accenture, Mumbai, India | Senior Programmer | Telekom Austria/Telecom platform development | Jan 2010 – July 2012

- Led modules of service provisioning and **Cramer OSS Engine**, and Guided 2 - 3 junior developers every sprint.
- Implemented automation for improved testing accuracy and eliminated manual testing across the project.

OpenStrides, Bangalore, India | Programmer | CRM Software/Way2Wealth | Aug 2009 – Jan 2010

- Development and Performance optimization of Java based inhouse CRM web applications using MVC framework

Research

University of Illinois at Chicago | Research Assistant | Jan 2013 – May 2013; Aug 2013 – May 2014

- Static Code analysis for execution tracing & instrumenting using Optimization Framework

Sarvajani College of Engineering & Technology | Published Paper - ZEOS: An OS from scratch | CSI, ETCT 2008